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Term Project

Implementation and Evaluation of a Lightweight Real-Time Detection Framework for Deauthentication Attacks in WPA3-SAE Networks

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1-1 Preface

Wireless networks have become an essential part of modern communication infrastructure and are widely used in homes, organizations, universities, industrial systems, and Internet of Things applications. As reliance on Wi-Fi networks continues to expand, there is an increasing need for protection mechanisms that preserve data confidentiality, integrity, and service availability, especially in environments where continuous connectivity is required.

WPA3 was introduced to address several weaknesses found in earlier generations of wireless security protocols. WPA3-Personal uses SAE (Simultaneous Authentication of Equals) for authentication and supports MFP (Management Frame Protection). Despite these improvements, the reference study indicates that certain deployment or implementation conditions may still allow attacks targeting management frames to have an impact. One of the most important examples is the deauthentication attack, which can disconnect a legitimate user from the access point and cause a denial-of-service condition [1].

This project implements and studies a lightweight mechanism for detecting deauthentication attacks in WPA3-SAE networks by analyzing the temporal sequence of wireless frames. Instead of relying only on the number of Deauthentication frames, the system monitors frames that appear immediately afterward, such as Association Response and EAPOL Message 1, to identify an inconsistent connection-state transition that may indicate a forged Deauthentication frame [1].

At the current stage, the project aims to build a practical test environment, capture and analyze network traffic, implement a timing-based detection algorithm, and record the resulting alerts. The system will later be evaluated using appropriate metrics such as detection rate, false-positive rate, and detection latency, without assuming results before the practical experiments are completed.

1-2 Abstract

Wi-Fi networks, including those based on WPA3-SAE, remain exposed to attacks that can affect connection availability. One important threat is the deauthentication attack, in which forged management frames are transmitted to force a legitimate client to disconnect from an access point. This project aims to implement and evaluate a lightweight real-time framework for detecting this type of attack using timing-based analysis of IEEE 802.11 frames.

The detection mechanism monitors a Deauthentication frame and then analyzes the subsequent frames within a 500 ms observation window. If the access point continues by transmitting an Association Response or EAPOL Message 1, this creates a protocol-state inconsistency that can be used as an indicator of a potential attack [1]. The project will be implemented in a Linux environment using Wireshark, Tshark, and Python for packet capture,

analysis, and alert logging, and the system will later be evaluated using detection rate, false-positive rate, and detection latency.

Keywords: WPA3, SAE, Deauthentication Attack, Wireless Network Security, Intrusion Detection, Timing Analysis, Wi-Fi.

1-3 Introduction

In recent years, the use of wireless networks has expanded significantly because of their ease of deployment and their ability to provide connectivity to a wide range of devices. However, the open nature of the wireless medium makes Wi-Fi networks vulnerable to several security threats, including eavesdropping, impersonation, frame injection, and denial-of-service attacks.

Wi-Fi security protocols have evolved from WEP to WPA, WPA2, and finally WPA3 in order to address weaknesses discovered in previous generations. WPA3 introduced several security improvements, including the use of SAE in WPA3-Personal networks, which provides a stronger authentication mechanism than earlier approaches, together with support for Management Frame Protection (MFP) [1].

Despite these improvements, WPA3 networks may still be exposed to Deauthentication attacks in some operating environments, particularly when management frame protection is not applied as intended. In the scenario studied, an attacker spoofs the MAC address of the access point and sends a forged Deauthentication frame to the client. This may cause the client to return to an unauthenticated and unassociated state, while the legitimate access point continues the connection process because it is unaware of the attack [1].

1-4 Scientific Problem

The scientific problem lies in the difficulty of distinguishing a legitimate Deauthentication frame from a forged one based only on the presence of the frame. An attacker can spoof the MAC address of the access point, making the forged message appear similar to a legitimate message in terms of its apparent source. In addition, relying only on the number of Deauthentication frames or on a fixed signature may produce false positives or fail to detect some cases. Therefore, the problem is to develop a lightweight real-time mechanism that uses the sequence and timing of IEEE 802.11 frames to distinguish normal connection termination from inconsistent behavior caused by a forged frame [1].

1-5 Project Objective

The main objective of the project is to implement and evaluate a lightweight real-time mechanism for detecting deauthentication attacks in WPA3-SAE networks using temporal correlation between wireless frames, and then test its ability to distinguish normal behavior from suspicious sequences without requiring computationally heavy machine-learning models. The specific objectives are:

- Study the operation of WPA3-SAE and the sequence of frames involved in authentication and connection establishment.
- Build a suitable test environment for capturing and analyzing Wi-Fi traffic.
- Implement a lightweight detection algorithm based on temporal correlation between Deauthentication frames and subsequent frames.

• Record suspicious events and alerts and present their information in a form suitable for analysis.

• Evaluate the system using metrics such as detection rate, false-positive rate, and detection latency.

1-6 Project Importance and Motivation

The importance of the project comes from the fact that deauthentication attacks target service availability, and even short interruptions may be significant in sensitive environments. A lightweight mechanism that does not require machine-learning training or large computational resources is also suitable for study and implementation in resource-limited practical environments. The reference study shows that temporal characteristics play an important role in distinguishing normal behavior from malicious behavior [1].

1-7 State of the Art

Recent research on Deauthentication attack detection follows several major directions: analyzing traffic, Reason Codes, and timestamps; monitoring management-frame sequences; applying machine-learning algorithms; and relying on Management Frame Protection as a preventive mechanism. The reference study shows that each direction has advantages and limitations in terms of accuracy, computational requirements, real-time operation, and compatibility with WPA3-SAE networks [1].

Study / Approach	Main Idea	Key Observation or Limitation
Noman et al. [2]	Python-based automated tool for detecting Deauthentication and Disassociation attacks in IEEE 802.11 networks.	Effective on Linux, with compatibility limitations on Windows.
Singh & Kumar [3]	Uses Radiotap information, Reason Codes, and MAC timestamps to reduce false positives.	Lightweight, but the parameters require further refinement to improve accuracy.
Saini et al. [4]	RIDS uses machine learning to detect attacks in WPA3 Enterprise environments.	High detection capability, but with greater computational and training-data requirements.
Dalal et al. [5]	Wireless intrusion detection for WPA3 networks based on monitoring the frame sequence exchanged between client and access point.	Frame-sequence monitoring provides an important comparison point for lightweight behavioral detection.
Reference IDR-DF [1]	Timing-based, behavior-oriented monitoring of Deauthentication frames and subsequent protocol-state transitions.	Real-time and interpretable without model training; the full recovery mechanism is outside the current implementation stage.

Based on this review, the current project adopts a lightweight protocol-behavior and timing-based approach because it does not require prior training and allows the detection decision to be interpreted directly. The core detection mechanism will be implemented first, while the complete recovery mechanism and more advanced extensions will remain for later project stages.

1-8 Proposed Solution Idea

The proposed solution passively monitors WPA3-SAE network traffic using a Wi-Fi interface operating in Monitor Mode. IEEE 802.11 frames are captured using Wireshark or Tshark, after which a selected set of features is extracted, including the Timestamp, MAC addresses, BSSID, frame type and subtype, Reason Code, and EAPOL message number.

When a Deauthentication frame is detected, the system records its information and opens a 500 ms observation window. If an Association Response or EAPOL Message 1 associated with the same AP/BSSID appears during this window, the sequence is considered an inconsistent protocol-state transition and is recorded as suspicious. If this pattern does not appear, the system continues monitoring without generating an alert [1].

The proposed practical implementation consists of a WPA3-SAE access point, a legitimate client, and a monitoring device equipped with a Wi-Fi interface that supports Monitor Mode, together with Ubuntu, Wireshark/Tshark, and Python. Python will be used to process packets, apply the detection rule, store results in logs and CSV files, and generate graphs for traffic analysis.

1-9 Project Block Diagram

The following diagram shows the flow of data from the wireless network environment to packet capture, feature extraction, detection, and finally logging and result visualization. The recovery mechanism is kept as a later stage because it is not part of the current implementation.

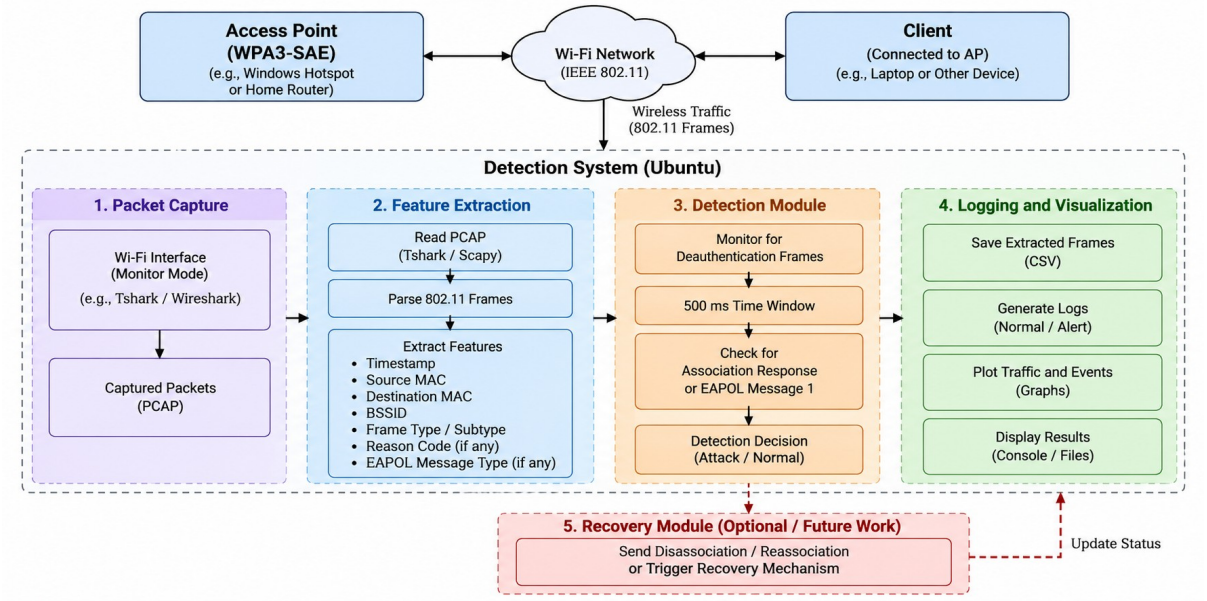


Figure 1: Block diagram of the proposed solution.

1-10 Detection Algorithm Flowchart

The flowchart illustrates the decision logic of the detector. The system continuously monitors frames. When a Deauthentication frame appears, it opens a 500 ms observation window and checks whether an Association Response or EAPOL Message 1 from the same AP/BSSID appears. The sequence is then classified according to the result of this check.

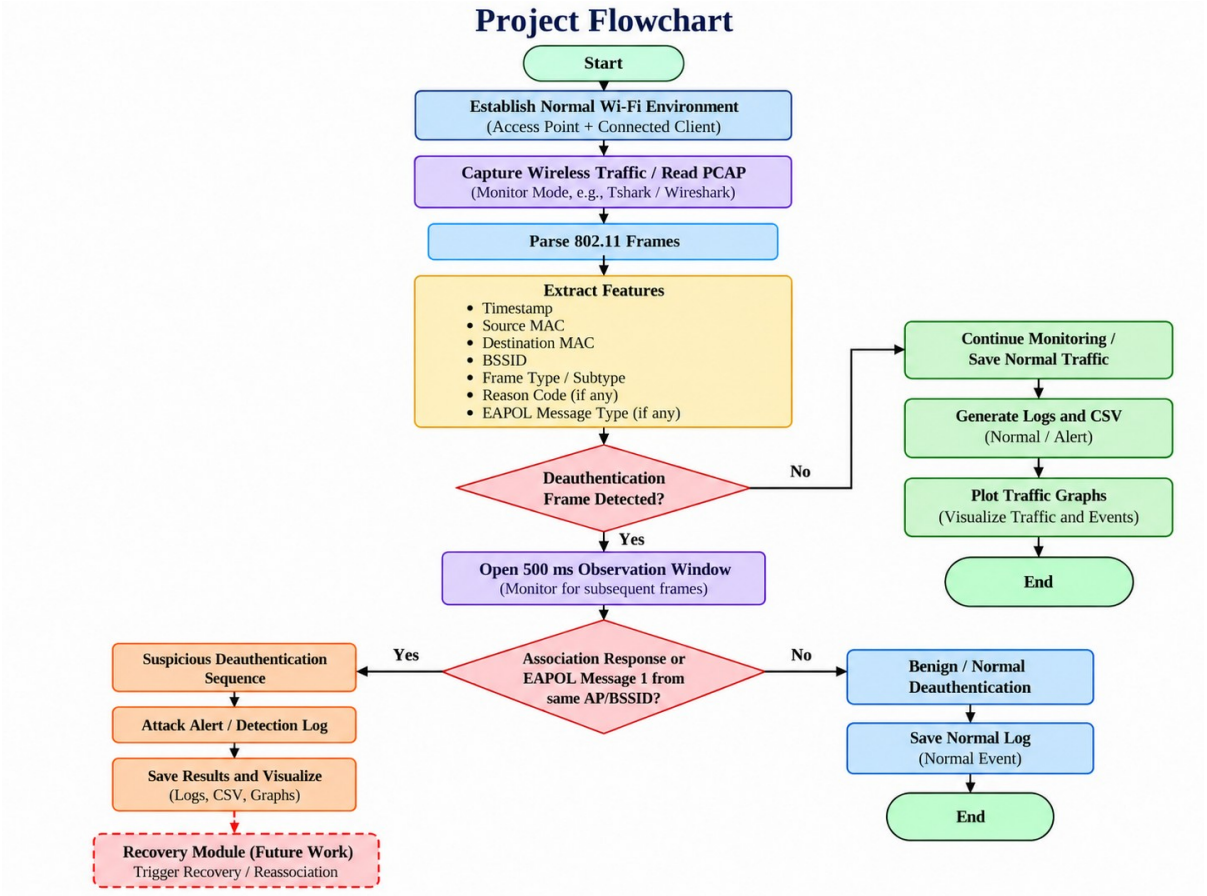


Figure 2: Flowchart of the proposed detection algorithm.

Chapter One: Theoretical Framework

This chapter presents the concepts, protocols, and theoretical elements that are directly used to build the project and explain its detection mechanism. Concepts that are not directly related to the practical implementation are intentionally omitted in order to keep the theoretical study focused on the project requirements.

1-1 Wireless Networks (Wi-Fi)

Wi-Fi is one of the most widely used wireless networking technologies and is based on the IEEE 802.11 family of standards for organizing communication between devices and access points. Communication in a Wi-Fi network depends on the exchange of different frames between the client and the Access Point (AP). This project focuses mainly on management frames involved in authentication, association, and deauthentication, together with EAPOL messages that appear during the key-establishment stage.

1-2 WPA3 Protocol

WPA3 represents a newer generation of Wi-Fi security protocols and was designed to address several weaknesses present in WPA2. In WPA3-Personal networks, SAE is used instead of directly relying on the traditional password mechanisms used in earlier generations, and WPA3 also supports Management Frame Protection (MFP). The reference study indicates that the actual level of protection may still depend on how network devices implement and configure these mechanisms [1].

1-3 SAE Mechanism

SAE stands for Simultaneous Authentication of Equals. It is an authentication mechanism used in WPA3-Personal that allows two parties to establish shared cryptographic information based on a shared password without transmitting the password directly. The connection process includes SAE message exchanges, followed by the Association stage and then messages related to the establishment of session keys. Understanding this sequence is important because the project detection mechanism determines whether the observed frame sequence is normal or inconsistent.

1-4 Management Frame Protection (MFP)

MFP stands for Management Frame Protection and is intended to protect selected IEEE 802.11 management frames against spoofing and manipulation. This protection is important because management frames control states such as connection establishment and termination. The reference study presents IDR-DF not as a replacement for MFP, but as an additional layer that monitors the temporal consistency of management-frame exchanges in environments where protection may be incompletely configured or implemented [1].

1-5 Deauthentication Attack

A Deauthentication attack targets the availability of wireless connections. In the studied scenario, the attacker spoofs the MAC address of the access point and sends a Deauthentication frame to the client in order to force it into an unassociated state. Meanwhile, the legitimate access point may continue the connection process because it did not send the forged frame. This inconsistency in the behavior of the two legitimate parties is the basis used by the detection algorithm [1].

1-6 Network Frames and Features Used in the Project

The project analyzes a limited set of information from captured packets in order to keep the system simple and lightweight. The main fields used in the reference study include [1]:

- Frame timestamp (Timestamp / frame.time).
- Sender and receiver MAC addresses (wlan.sa / wlan.ra).
- Access-point BSSID (wlan.bssid).
- Frame type and subtype (wlan.fc.type / wlan.fc.subtype).
- Reason Code in Deauthentication and Disassociation frames.
- EAPOL message number during the handshake stages.

1-7 Timing-Based Detection

The basic detection algorithm relies on Timing-Based Anomaly Detection. When a Deauthentication frame is detected, the system records the sender address, receiver address, and timestamp, and then monitors network traffic for 500 ms. If an Association Response or EAPOL Message 1 associated with the same access point appears during this interval, this represents an inconsistent connection-state transition and is classified as potentially malicious behavior [1].

The 500 ms window was selected empirically in the reference study after observing that forged Deauthentication attacks are followed quickly by attempts to continue the handshake, while the same pattern does not appear in normal connection termination. Therefore, the basic rule is not “the presence of a Deauthentication frame means an attack,” but rather “a Deauthentication frame followed by an inconsistent continuation of the handshake within a short time window indicates suspicious behavior” [1].

1-8 Connection Sequence Monitored by the Algorithm

In a normal connection, the client sends an Association Request, the access point returns an Association Response, and EAPOL messages then begin the key-establishment process. During an attack, however, the client may receive a forged Deauthentication frame while the legitimate access point continues sending an Association Response or EAPOL Message 1. Therefore, the system monitors the relationship and timing between these frames instead of relying on a single field or a fixed packet count.

References

[1] A. Btoush, A. Abadleh, A. A. Alkasasbeh, W. Abdallah, A. I. Zreikat, and N. Mostafa, “A Lightweight Real-Time Detection and Recovery Framework for Deauthentication Attacks in WPA3-SAE Networks,” International Journal of Software Engineering and Knowledge Engineering, 2026, doi: 10.1142/S0218194026500105.

[2] H. A. Noman, S. M. Abdullah, and H. I. Mohammed, “An automated approach to detect deauthentication and disassociation DoS attacks on wireless 802.11 networks,” International Journal of Computer Science Issues, vol. 12, no. 4, p. 107, 2015.

[3] R. Singh and S. Kumar, “A light weight solution for detecting de-authentication attack,” International Journal of Network Security & Its Applications, vol. 11, pp. 15–26, 2019.

[4] R. Saini, D. Halder, and A. M. Baswade, “RIDS: Real-time intrusion detection system for WPA3 enabled enterprise networks,” in Proc. IEEE GLOBECOM, 2022, pp. 43–48.

[5] N. Dalal, N. Akhtar, A. Gupta, N. Karamchandani, G. S. Kasbekar, and J. Parekh, “A wireless intrusion detection system for 802.11 WPA3 networks,” in Proc. COMSNETS, 2022, pp. 384–392.